Question Paper

Exam Date & Time: 07-Jan-2023 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

SEVENTH SEMESTER B.TECH END SEMESTER MAKEUP EXAMINATIONS, JAN 2023

Multi Sensor Data Fusion [ICE 4057]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions. Section Duration: 180 mins

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) What is sensor uncertainty? Describe different types of sensor uncertainty. [CO1, PO1, BL2] (3)

A)

B) Describe serial and parallel fusion networks with necessary diagrams. [CO1, PO1, BL2] (3)

C) Given two time series, P = (1.7, 3.6, 6.3, 2.7, 8.3, 6.9) and Q = (2.8, 6.4, 5.2, 1.1, 7.4, 8.3), calculate cummulative matrix D and also optimal warping path using dynamic programming. [CO2, PO2, PO3, BL4] (4)

2) What is the requirement of semantic alignment? (5)

A) Let X= (x1, x2, x3, x4, x5)^T denote an input vector. By partitioning X using two different clustering algorithms {A, B}, resulting identity vectors are: A= (α2 α1 α3 α3 α1), B= (β1 β2 β2 β3 β3). Write the corresponding two co-association matrices and the mean co-association matrix. [CO2, PO2, PO3, BL3]

B) Describe 4 scales of measurement. [CO2, PO2, PO3,BL2] (3)

C) Explain Dasarthy's Input/output Data fusion model [CO4, PO1, BL2] (2)

3) Given data for five people in Table. Each person vector has a Height, Score on some test, and Age. Determine the Mahalanobis distance of another person v = (59, 605, 49) from the set of data.[CO3, PO1, PO2, PO3, PO4, BL3] (4)

A)

X,Height	Y ,Score	Z,Age
74	644	47
63	556	32
71	473	33
65	529	46
69	620	55
74	651	59

B) For the given values of Y= (13, 3, 4, 12, 14, 10, 5, 1, 8, 2, 7, 9, 11, 6, 18), form a binary search tree [CO3, PO1, PO2, PO3, PO4, BL3] (2)

C) Explain Waterfall model of fusion process with an example [CO4, PO1, BL2] (4)

4) With neat figure, describe different levels of JDL fusion framework [CO4, PO1, BL2] (4)

A)			
B)		Explain Omnibus model with an example. [CO4, PO1, BL2]	(4)
C)		Describe the Luo-Kay data fusion framework. [CO4, PO1, BL2]	(2)
5)		Estimate the temperature of a vessel using the Kalman filtering technique for the following data:	(5)
	A)	True temperature = 72 Initial Estimate=68 Initial error in estimate=2 Error in measurement =4 Measurements= {75,71,70,74} [CO5, PO1, BL3]	
	B)	What is Bayesian filtering? Classify the types of Bayesian filters[CO5, PO1, BL2]	(3)
	C)	Explain the need for data filtering. [CO5, PO1, BL2]	(2)

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